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Introduction

This report is based on an employee dataset that contains general information about workers in a company. The dataset includes details such as department, job title, age, gender, hire date, and employee status.

In general, this data helps give a clear picture of the employees in the company. It can be used to understand how employees are distributed.

By analyzing this dataset, it becomes easier to understand the workforce and draw helpful conclusions about the company.

Dataset Description

The dataset contains information about employees in a company. Each row represents one employee, and each column represents a specific detail about that employee. The data includes personal information such as age, gender and work-related information such as age department, job title, hire date, and employment status.

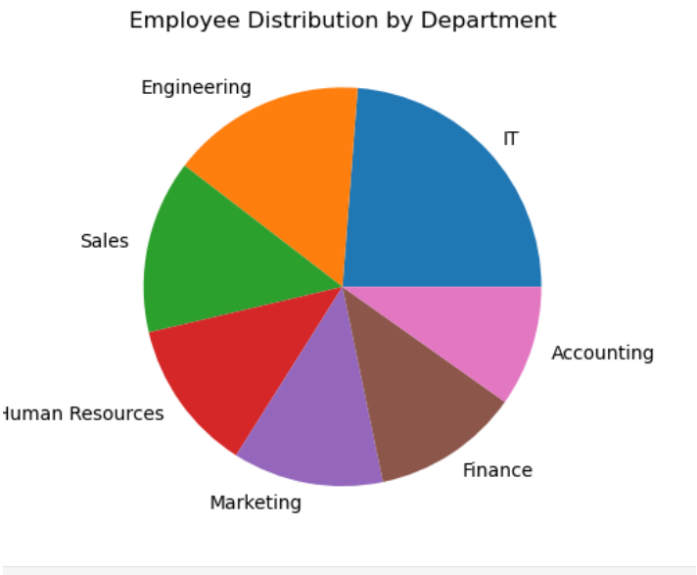
This dataset is useful because it helps organize employee information in one place. It also makes it easier to compare employees across different departments and understand the overall structure of the company.

Data Cleaning and Preparation

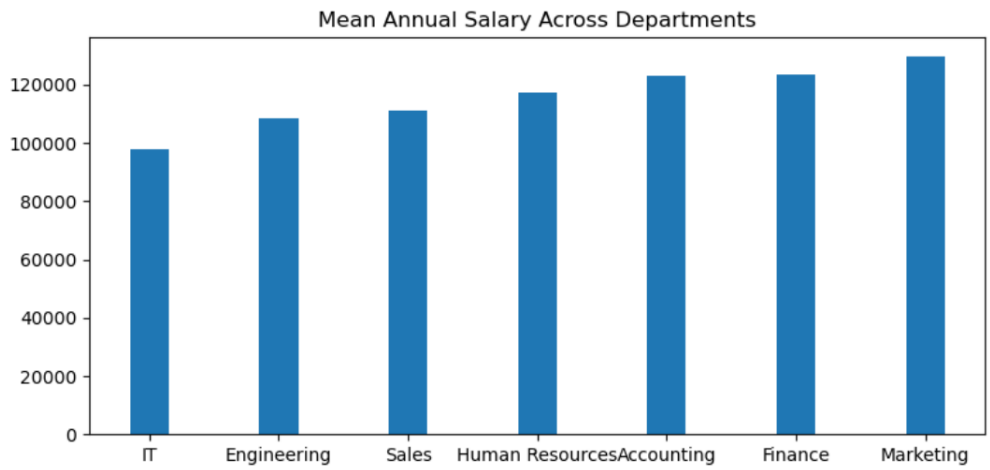
Data preparation was an important step before starting the analysis. It included reviewing the dataset, organizing the data in a suitable format, and paying special attention to employees with missing information. This was necessary because missing or unclear values can affect the accuracy and quality of the results and may lead to unclear or misleading conclusions if they are ignored. For this reason, the dataset was checked carefully, and missing information was handled based on its type and importance. In some cases, unnecessary missing data was removed, while in other cases the available information was kept and interpreted carefully. After that, the dataset became more organized and ready for analysis. results.

Analysis and Results

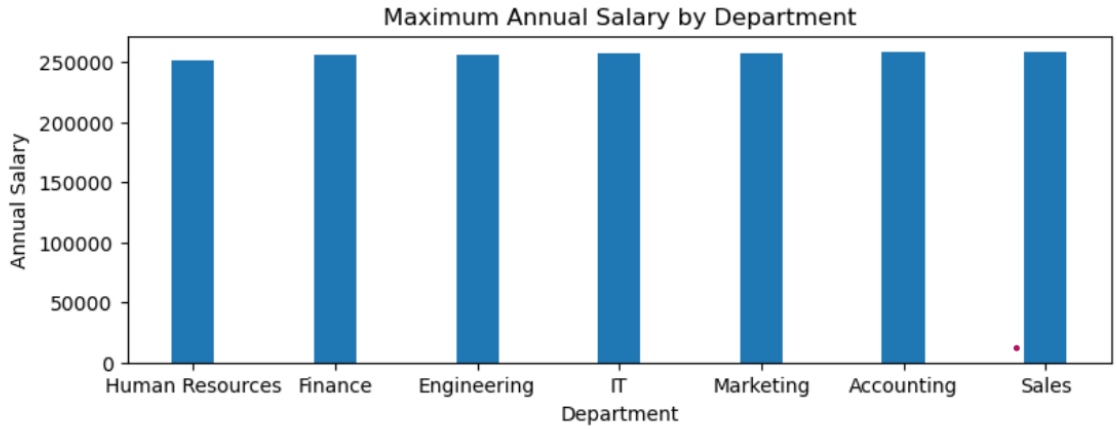
After preparing the dataset, the analysis was carried out to better understand the employees and the overall workforce structure. The data was examined from different angles, such as department distribution, gender, age, and employee status. This helped reveal clear patterns and important information about the company. The results made it easier to identify differences between departments, understand employee distribution, and notice areas that may need more attention.



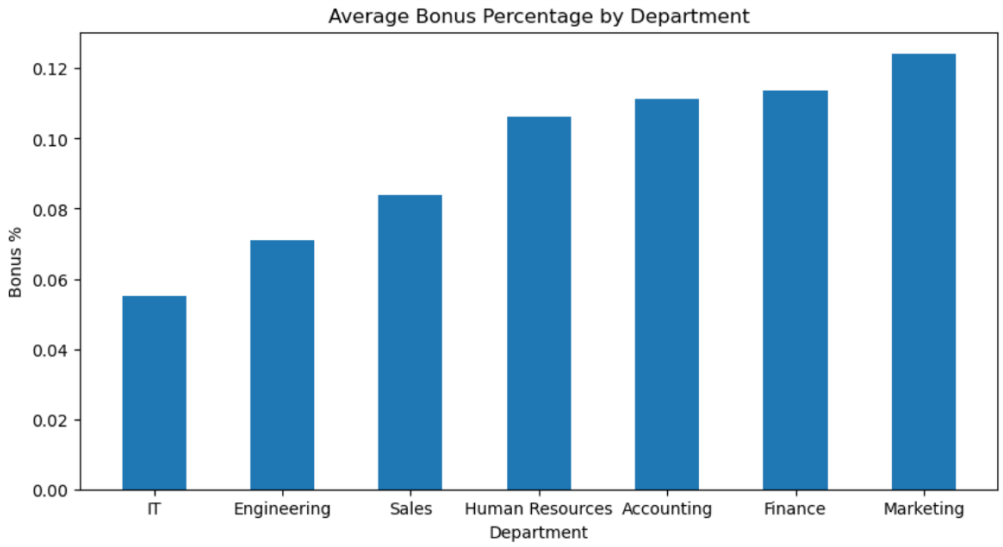
This chart shows the distribution of employees across different departments. The results indicate that IT has the largest share of employees, followed by Engineering and Sales. In contrast, Accounting has the smallest share. Overall, the chart shows that employee distribution is not equal across departments, with some departments having more employees than others.



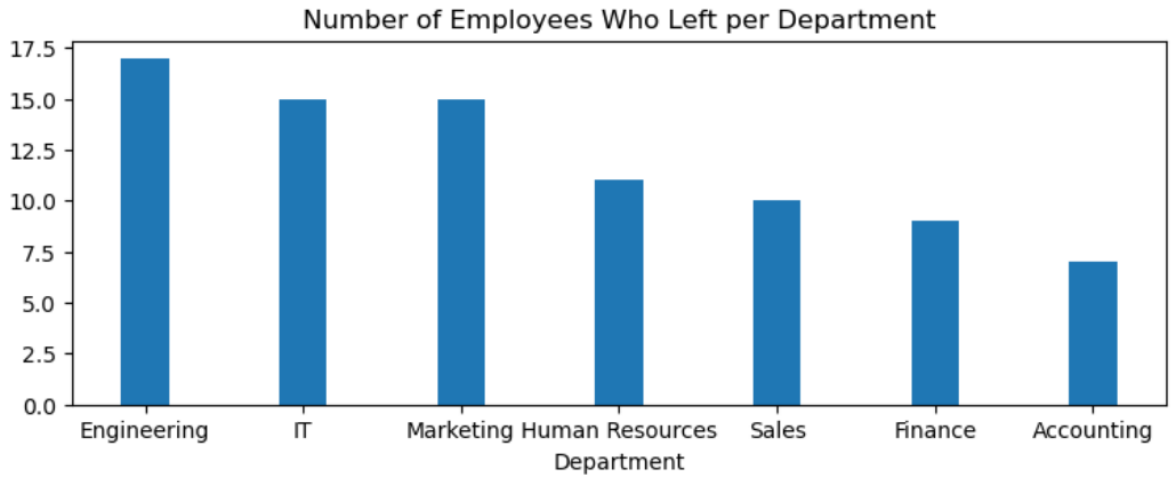
This chart shows the mean annual salary in each department. It compares the average salary of employees across departments. The results show that Marketing has the highest average salary, while IT has the lowest. This means that salary levels differ from one department to another.



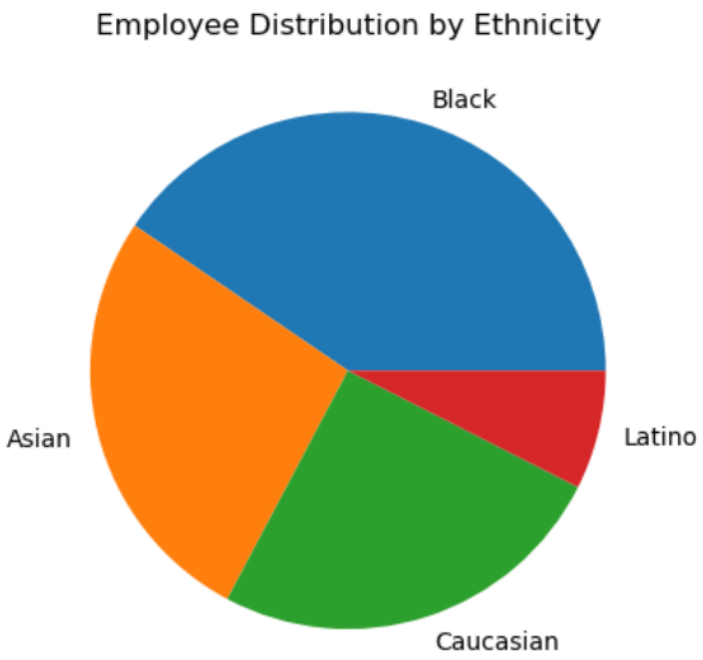
This chart shows the highest annual salary recorded in each department. The results indicate that the maximum salaries are relatively close across all departments, with only small differences between them. Among the departments, Sales has the highest maximum annual salary, while Human Resources has the lowest maximum value. Overall, this suggests that the top salary range in the company is fairly similar across departments.



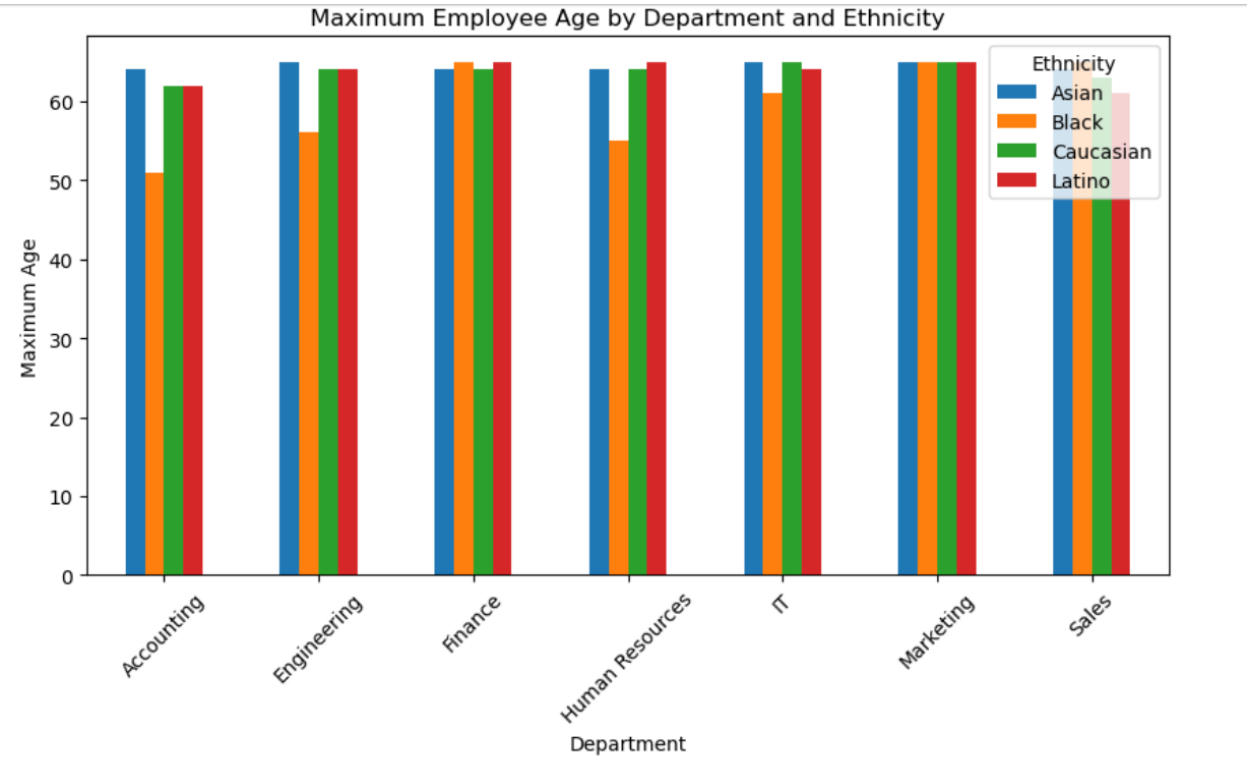
This chart shows the average bonus percentage across departments. The results indicate that bonus rates differ from one department to another. Marketing has the highest average bonus percentage, followed by Finance and Accounting, while IT has the lowest average bonus percentage. Overall, this suggests that employees in some departments receive higher bonus rates than others.



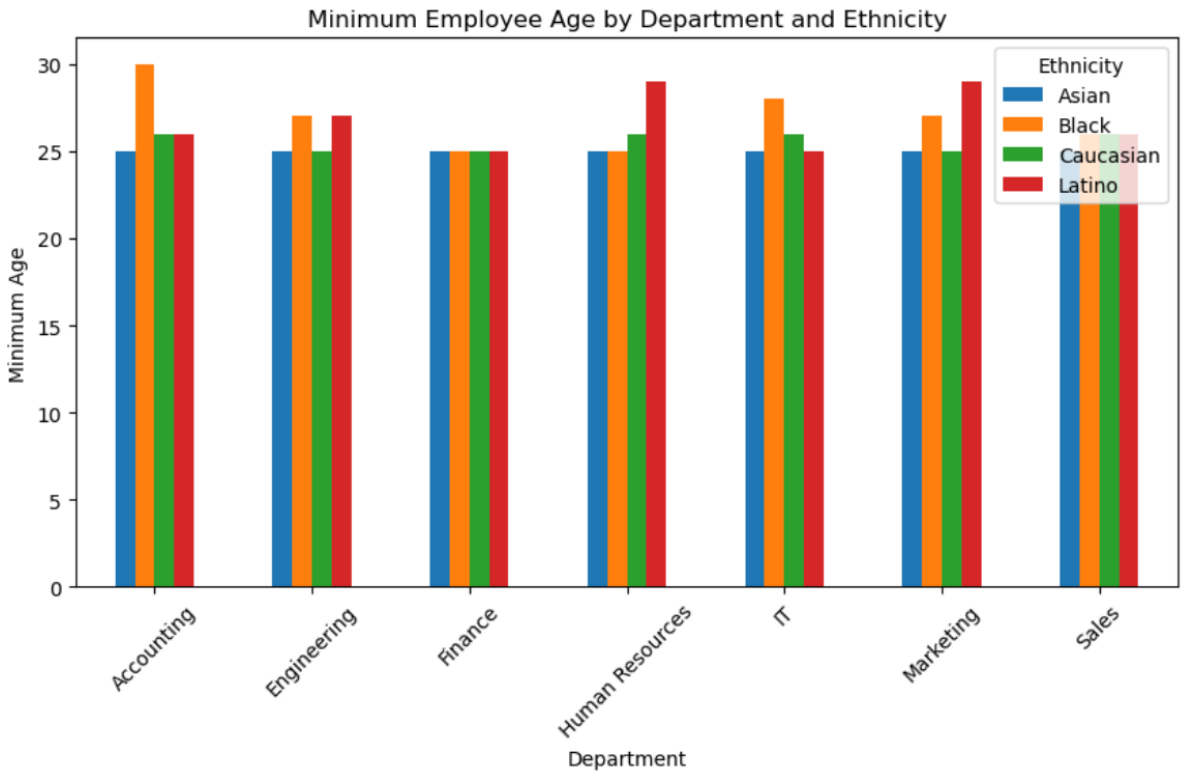
This chart shows the number of employees who left the company in each department. The results indicate that Engineering has the highest number of employees who left, followed by IT and Marketing. In contrast, Accounting has the lowest number of employee exits. Overall, the chart suggests that employee turnover is higher in some departments than in others, which may indicate differences in retention across the company.



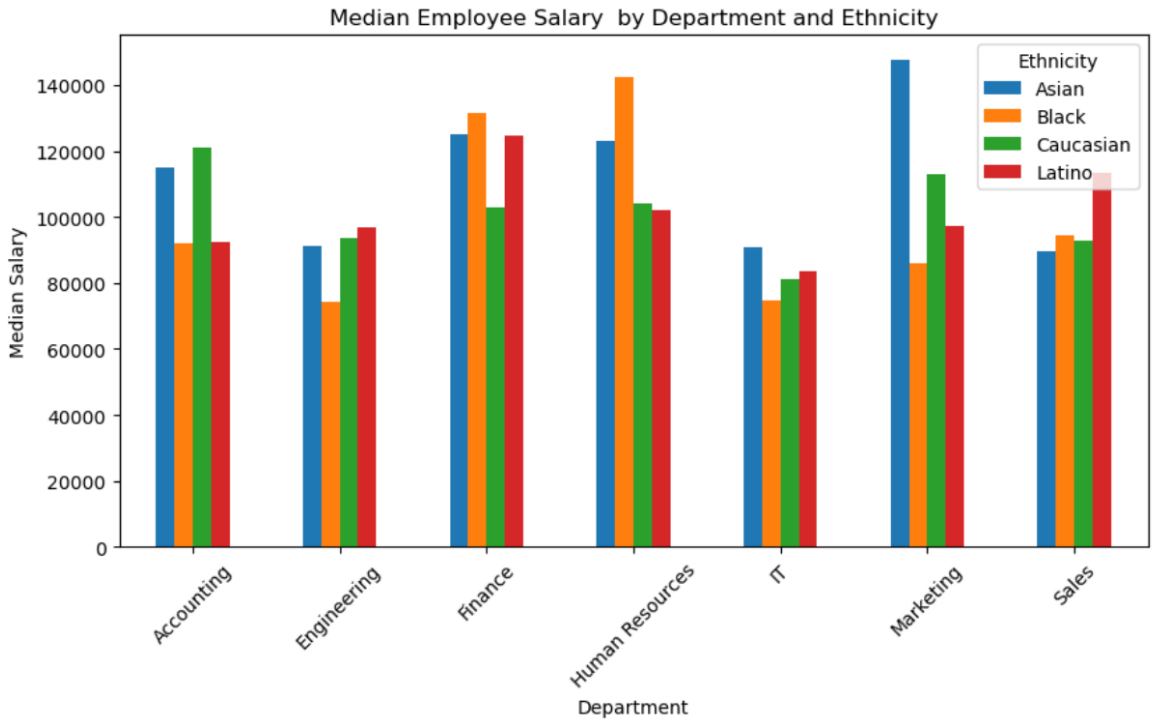
This chart presents employee distribution by ethnicity. It shows that Black employees make up the largest part of the workforce, while Latino employees make up the smallest. Asian and Caucasian groups are also clearly represented in the company.



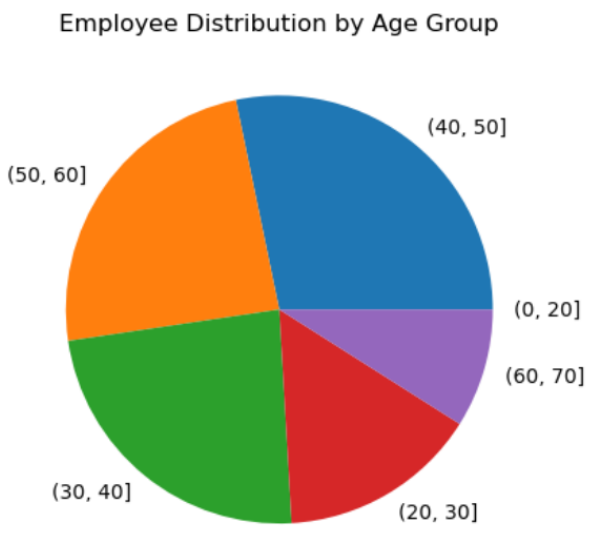
The analysis of maximum age by department and ethnicity shows that the highest ages are generally close across most groups, with values mostly ranging from 61 to 65 years. This suggests that there is no major difference in the upper age range between departments and ethnicities. Although Asian employees appear among the oldest groups in some departments, this pattern is not consistent across all departments, since other ethnic groups also show equal or higher maximum ages in several cases.



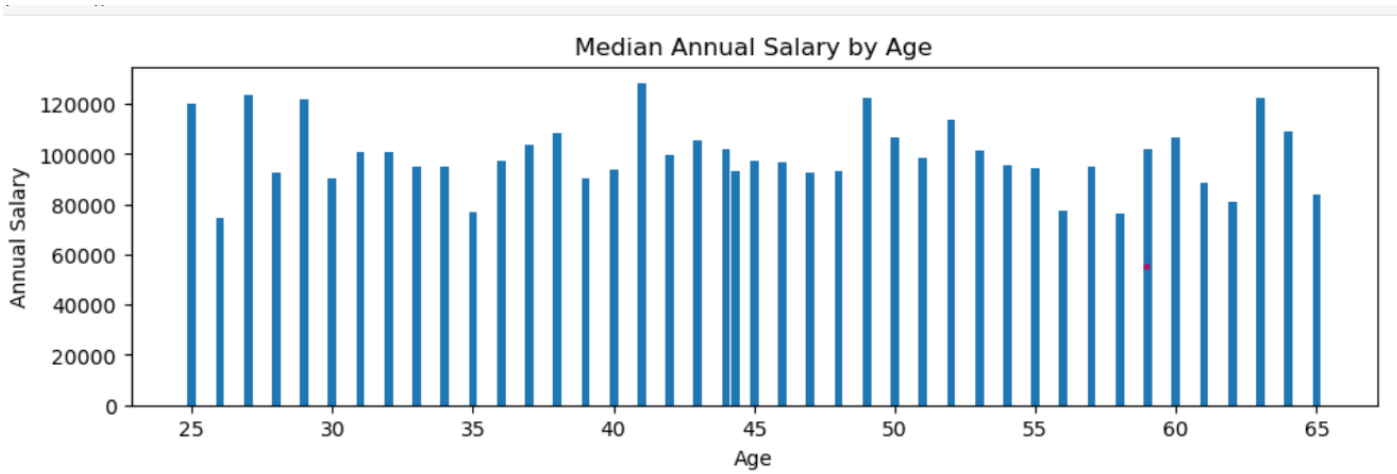
This analysis shows the minimum employee age in each department based on ethnicity. The results indicate that the youngest ages are generally close across most groups, mainly ranging between 25 and 30 years. In many departments, the Asian group has the lowest minimum age, often around 25 years, while some other ethnic groups show slightly higher minimum ages. Overall, the differences are small, which suggests that the lower age range is relatively similar across departments and ethnicities.



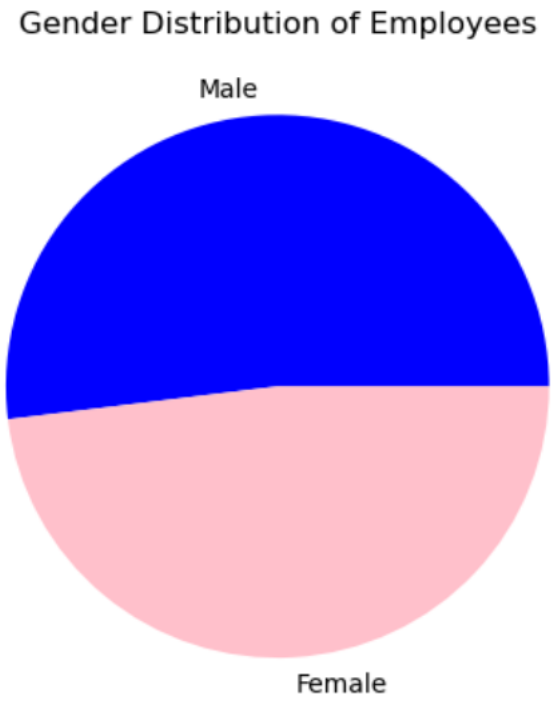
The median salary analysis shows clear differences across department and ethnicity groups. Marketing has the highest median salary for Asian employees, while Human Resources has the highest median salary for Black employees. Finance also shows relatively high median salaries for several groups. In contrast, IT and Engineering generally have lower median salary values. This suggests that both department and ethnicity are associated with differences in salary levels within the dataset.



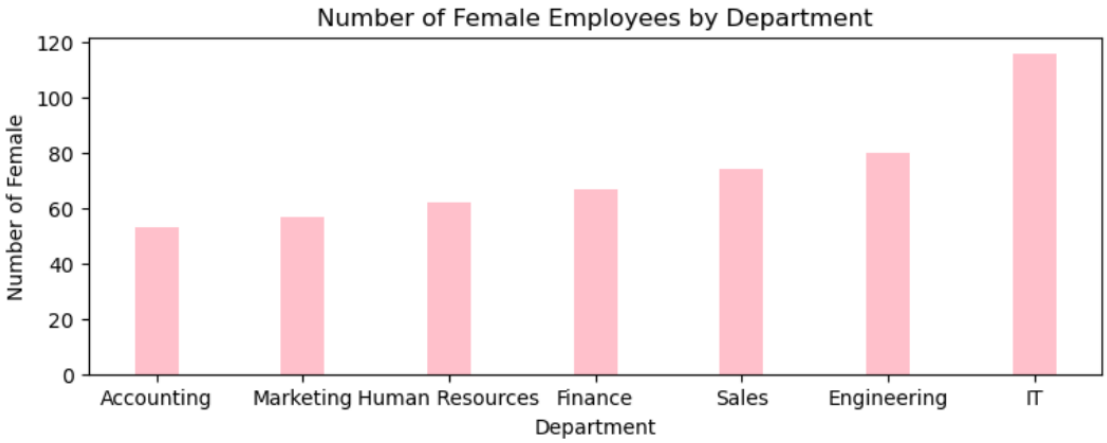
This chart shows the distribution of employees across different age groups. The results indicate that most employees fall within the age ranges of 40–50, 50–60, and 30–40, while fewer employees are found in the younger and older age groups. The 0–20 group appears to have little or no representation, which suggests that the workforce is mainly concentrated in the middle-age categories. Overall, the chart gives a clear view of the age structure of employees in the company.



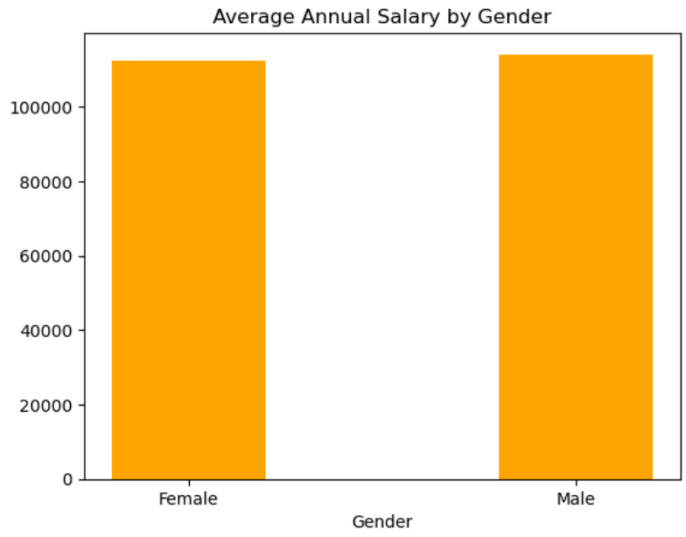
This analysis shows the median annual salary for each age group. The results indicate that salary does not increase in a perfectly regular way with age, but instead changes from one age to another. Some age groups, such as 41, 49, and 63, show relatively high median salaries, while others, such as 26, 35, 56, 58, and 65, show lower median values. Overall, the chart suggests that median salary varies across age groups, which means age alone does not explain salary differences in a simple way.



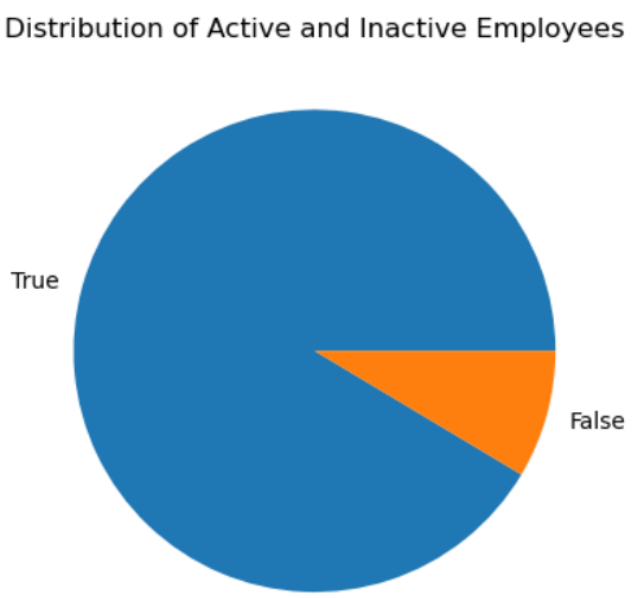
This chart shows the distribution of employees by gender. The results indicate that the number of male and female employees is nearly equal, which suggests a balanced gender representation in the company. This means that the workforce is distributed quite evenly between the two groups.



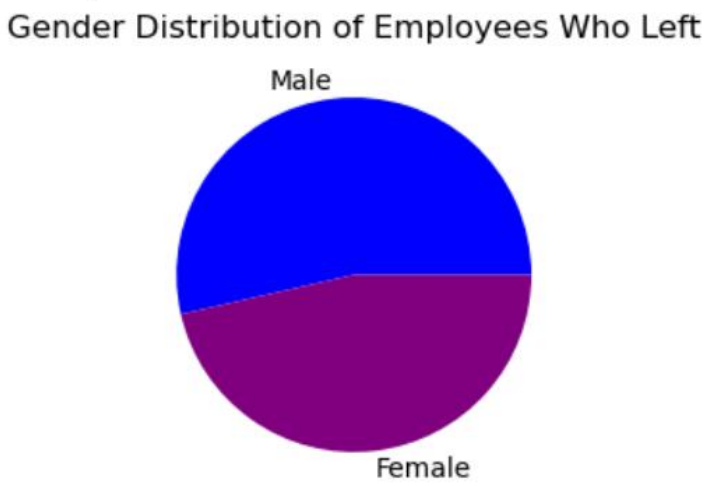
This chart shows the number of female employees in each department. The results indicate that IT has the highest number of female employees, followed by Engineering and Sales. On the other hand, Accounting has the lowest number of female employees among the departments shown. Overall, the chart highlights differences in the distribution of female employees across departments.



This chart presents the average annual salary by gender. It shows that male and female employees have nearly similar average salaries, with males being slightly higher. This suggests only a small difference between the two groups.

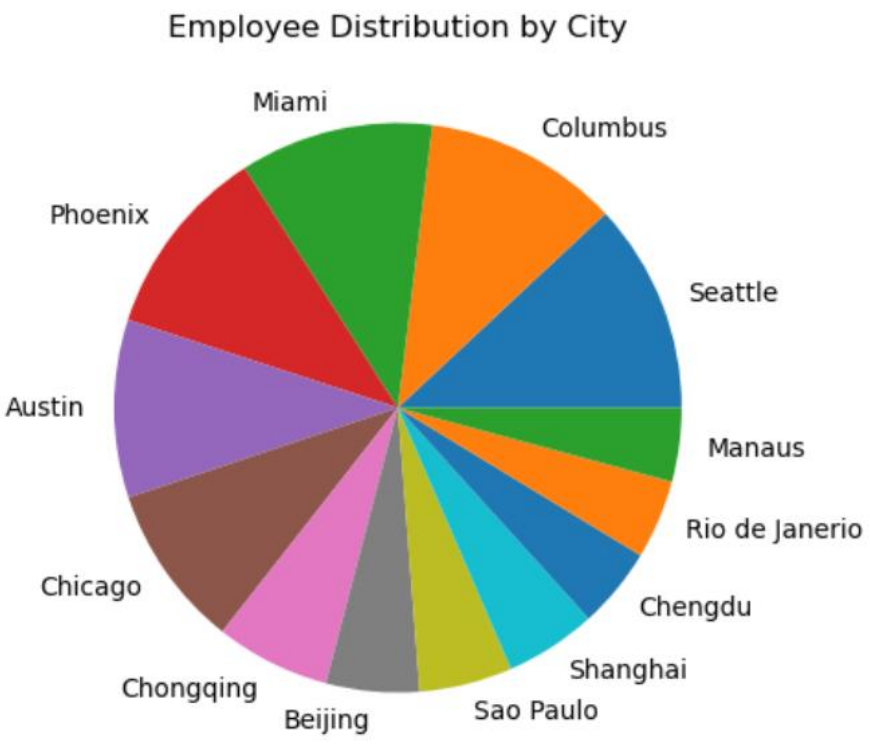


This chart presents employee status as active and inactive. It shows that most employees are active, while only a smaller group is inactive. This gives a general idea about the current workforce situation in the company.

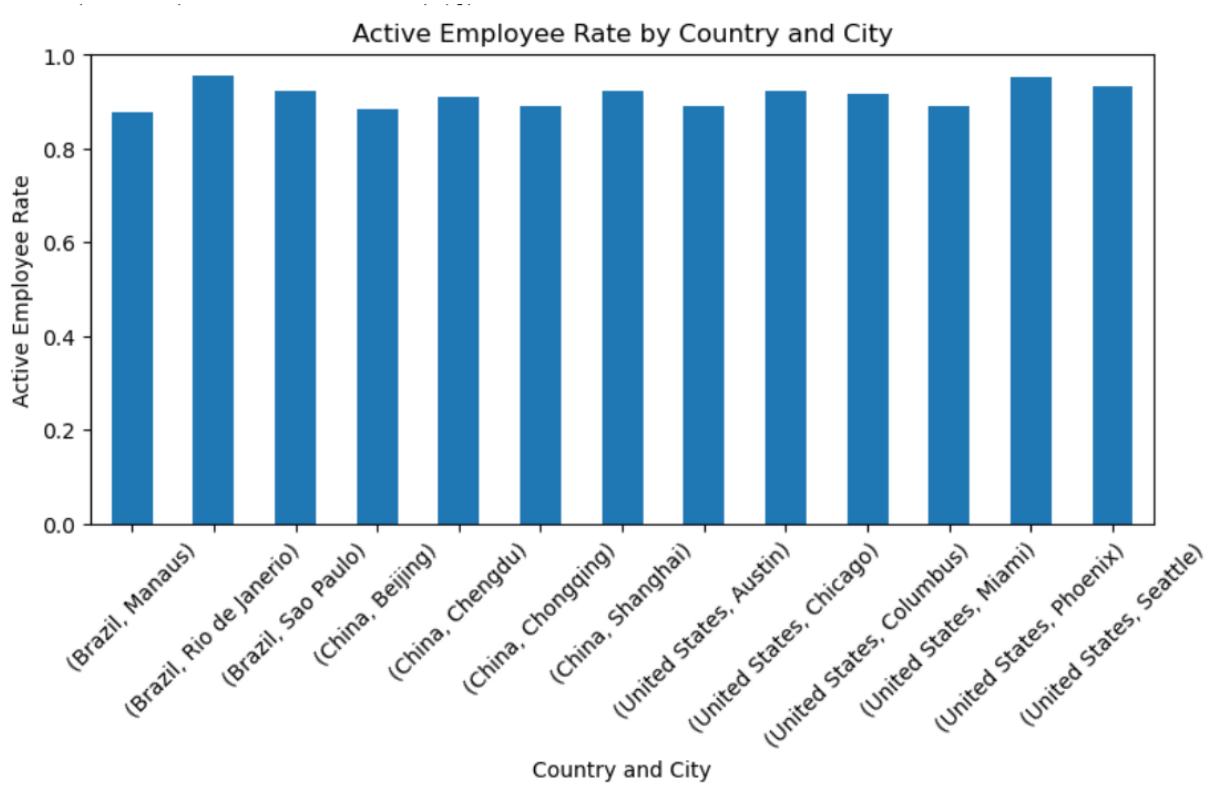


This chart presents the gender distribution of employees who left the company. It shows that males and females left in almost equal numbers, although males are slightly more. This means there is no strong gap between the two groups.

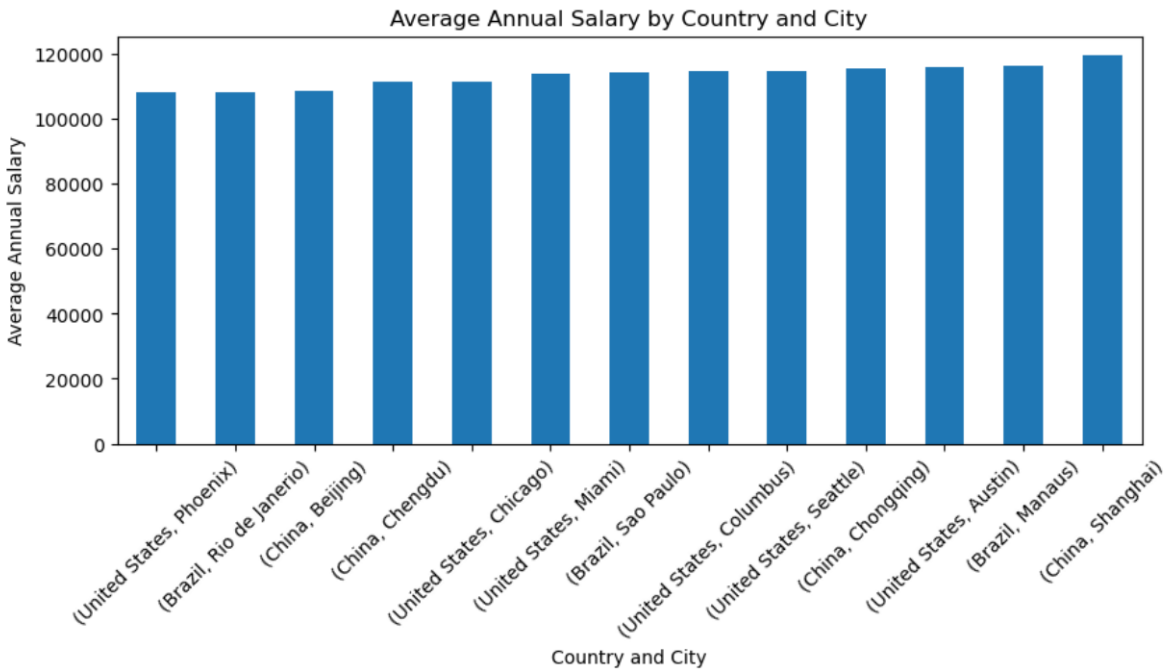
This chart shows the distribution of employees by country. The results indicate that the United States has the largest share of employees in the dataset, followed by China, while Brazil has the smallest share. This suggests that the company's workforce is mainly concentrated in the United States, with a smaller presence in China and Brazil.



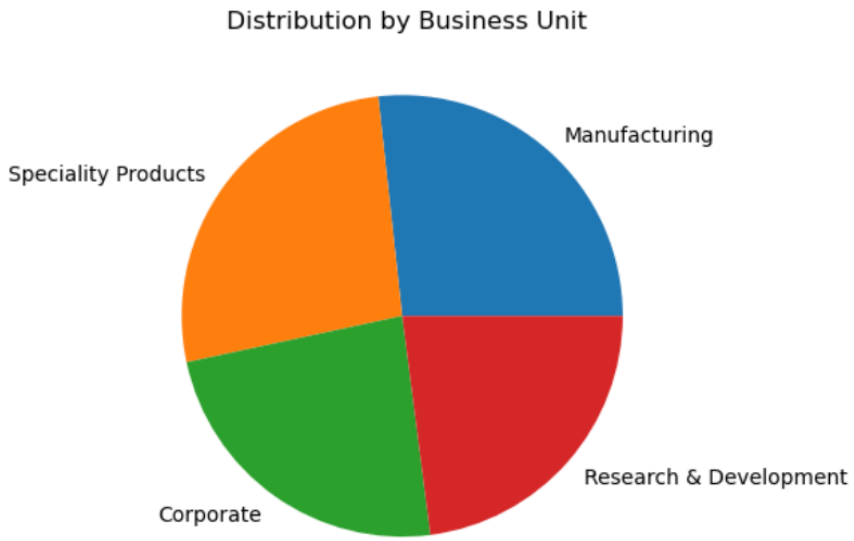
This chart shows the distribution of employees by city. The results indicate that employees are spread across several cities, but the distribution is not equal. Some cities, such as Seattle, Columbus, Miami, and Phoenix, have larger shares, while cities such as Manaus, Rio de Janeiro, Chengdu, and Shanghai have smaller shares. Overall, the chart provides a detailed view of how the company's workforce is distributed across locations.



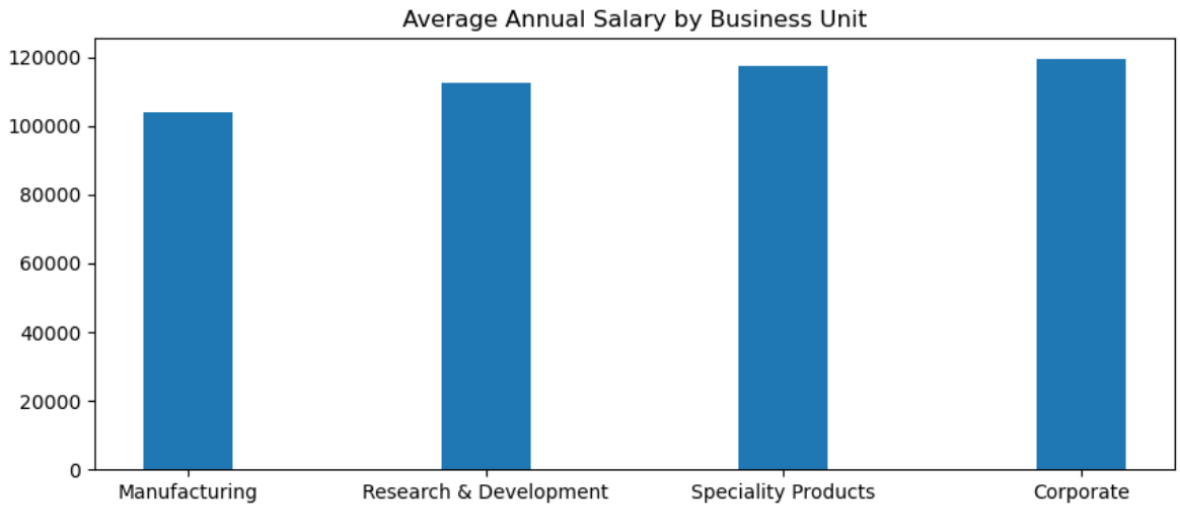
This chart shows the active employee rate by country and city. The results indicate that the proportion of active employees is high across all locations, with values generally close to 90% or more. Some cities, such as Rio de Janeiro and Miami, appear to have slightly higher active employee rates, while Manaus and Beijing show comparatively lower rates. Overall, the chart suggests that most employees remain active regardless of location, with only small differences between cities.



This chart shows the average annual salary by country and city. The results indicate that salary levels are relatively close across all locations, with only small differences between them. Shanghai appears to have the highest average annual salary, while Phoenix and Rio de Janeiro show the lowest average values among the cities displayed. Overall, the chart suggests that average salary is fairly consistent across locations, although some cities have slightly higher salary levels than others.



This chart shows the distribution of employees by business unit. The results indicate that employees are fairly balanced across the four business units, with only small differences between them. Manufacturing and Speciality Products appear to have slightly larger shares, while Research & Development has the smallest share. Overall, the chart suggests that the workforce is distributed relatively evenly across the business units.



This chart shows the average annual salary by business unit. The results indicate that salary levels differ slightly across the four business units. Corporate has the highest average annual salary, followed by Speciality Products and Research & Development, while Manufacturing has the lowest average salary. Overall, the differences are not very large, which suggests that average salary is relatively close across business units.

Analysis and Results

After preparing the dataset, several analyses were performed to understand the company's workforce. The results showed that salary and bonus levels vary slightly across departments, with Marketing and Finance generally having higher values, while IT has lower ones.

The analysis also revealed differences in salary across ethnicity and age groups, although these differences were not always consistent. In terms of workforce composition, gender distribution was balanced, and most employees were active, with fewer employees who left the company. Employee exits were higher in some

departments, such as Engineering, compared to others like Accounting.

The analysis of ethnicity and age groups showed that employees are mainly concentrated in certain groups, especially middle age ranges. From a geographic perspective, most employees are located in the United States, with smaller groups in China and Brazil. Differences between cities were present but not very large, and most locations showed a high rate of active employees.

Overall, the results highlight that while many aspects of the workforce are balanced, there are still noticeable differences between departments, employee groups, and locations.

Main Insights

The main insight is that the workforce is generally balanced, but there are differences between departments. Some departments have higher salaries and bonuses, while others have lower values. Employee exits are higher in certain departments. Most employees are active, and the workforce is balanced in gender but varies in age, ethnicity, and location.

Conclusion

In conclusion, the analysis helped provide a clear understanding of the company's workforce. The results showed that while many aspects such as gender distribution and employee activity are balanced, there are still differences in salary, bonuses, and employee exits across departments and groups. The analysis also highlighted how employees are distributed by age, ethnicity, and location. Overall, the dataset helped turn raw employee information into meaningful insights about the organization.

Recommendations

Based on the analysis, it is recommended to review departments with higher employee exits, such as Engineering, to improve employee retention. It is also important to ensure fairness in salary and bonus distribution across departments and employee groups. Additionally, the company can use these insights to better understand workforce distribution and support future planning and decision-making.